

Globalmed Project: Zero-Trust Architecture

Remediating High-Risk PII Exposure through Secure Data Engineering, RBAC,
and Executive Compliance Monitoring.

The Security Liability

Critical Vulnerability Detected

Internal audits at Globalmed revealed a severe structural flaw: sensitive patient data (PII) was globally accessible to unauthorized internal staff.

This fundamental violation of least-privilege principles presented immediate, high-severity compliance risks under HIPAA and GDPR regulatory frameworks.



Core Strategic Pillars



Data Governance

Implemented SQL-based Dynamic Data Masking (DDM) to obfuscate sensitive PII seamlessly, ensuring protection without breaking operational pipelines or data utility.



Security Engineering

Engineered a granular Role-Based Access Control (RBAC) model mapping to centralized identity management, enforcing strict least-privilege access across all data tiers.



Executive Reporting

Deployed an integrated Power BI compliance dashboard utilizing service principal authentication to securely visualize risk exposure and posture for C-suite stakeholders.



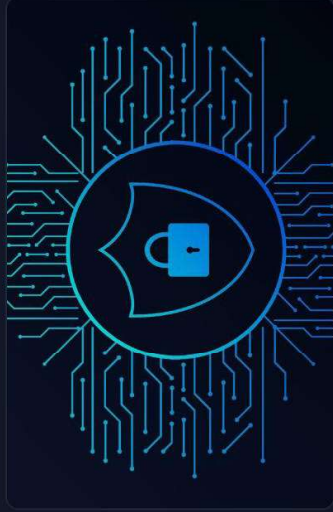
Architecture & Implementation

Building a scalable, enterprise-grade data security framework.

RBAC & Masking Matrix

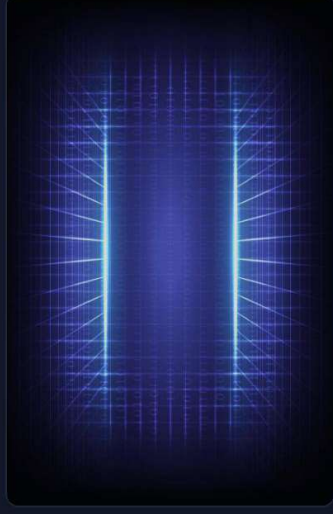
SECURITY CONTEXT	ACCESS LEVEL	EMAIL / SSN / CC VIEW	OPERATIONAL FUNCTION
Database Owner	System Admin	Unmasked	Full visibility prior to policy application.
Data Analyst	Restricted	Dynamically Masked	Authorized for operational aggregations without exposing raw PII.
Compliance Officer	Privileged	Unmasked	Granted explicit UNMASK permissions for formal auditing requirements.

Technical Stack Integration



Azure SQL Engine

Serves as the core enforcement point for Dynamic Data Masking and local RBAC policies.



Entra ID Integration

Enterprise scalability achieved by mapping SQL roles directly to Azure Active Directory security groups.



Power BI Analytics

Secure visualization tier using identity-based authentication to respect underlying SQL masking rules.

Enterprise Business Impact

100%

PII Exposure Mitigation

Zero Operational Downtime

Successfully eliminated unauthorized access to sensitive healthcare records across the global analytics environment.

- ✓ Preempted severe HIPAA/GDPR non-compliance fines.
- ✓ Maintained full data utility for operational teams.
- ✓ Automated future onboarding via Entra ID framework.

Questions?

Globalmed Security Engineering Showcase

Privacy & Compliance Architecture Review